Sampling-based Certified Planning with Graphs of Convex Sets

Peng Xie and Amr Alanwar

Abstract—**Planners on graphs of convex sets return trajectories that are collision-free by construction, provided the convex regions are collision-free. The region generator only promises that property probabilistically, and no planner in the family verifies it. We report the first measurement of what the gap costs. On a scaled 14-DOF bimanual library, 3.2% of interface samples are in collision, and a search-based GCS planner (GCS*) turns that volume error into a 62% answer error: 18 of 29 pick-and-place queries return trajectories that drive the arms through the shelves, up to $91\,$mm deep, reported as successes. Repairing the library does not work; a ten times stricter acceptance contract, sums-of-squares certified regions, and uniform margins each destroy the connectivity planning needs before they deliver soundness. We instead build a planner that certifies its answers. It samples the overlaps and shared faces of the decomposition, prunes with an admissible informed bound, and verifies the one candidate each search round proposes, continuously, by a chain of clearance certificate balls with no resolution parameter; failures are repaired with local in-region detours, and the convex polish is re-verified. Head-to-head on all 29 task queries it delivers zero invalid answers against 21 for the reference, reaches its first certified answer in $0.11\,$s against $1.59\,$s for the reference's unverified one, and reproduces the reference optimum exactly on every query whose reference answer is physically valid.**

I. Introduction

Planners built on graphs of convex sets (GCS) return trajectories that are collision-free by construction: the trajectory stays inside a library of convex regions, and the regions are collision-free. The second half of that sentence is an assumption inherited from the region generator. The original formulation [5] and its search-based descendants [8] take it as given, and neither re-examines the trajectory returned.

The generators that produce these libraries at scale make a probabilistic promise: after a batch of random samples inside a candidate region comes back clean, the region is accepted, and its colliding volume is small with high confidence [3]. Small is not zero. On a 14-DOF bimanual library built at default settings, 3.2% of the samples we draw on region overlaps are in collision. Feeding that library to GCS* [8] turns a 3.2% model error into a 62% answer error: 18 of 29 pick-and-place queries return trajectories that drive the arms through the shelves, by $39\,$mm at the median and $91\,$mm at the worst, for up to half the length of the path. It reports success on every one.

A planner is an optimizer, and that is why the error grows on the way through. It searches for the cheapest trajectory the

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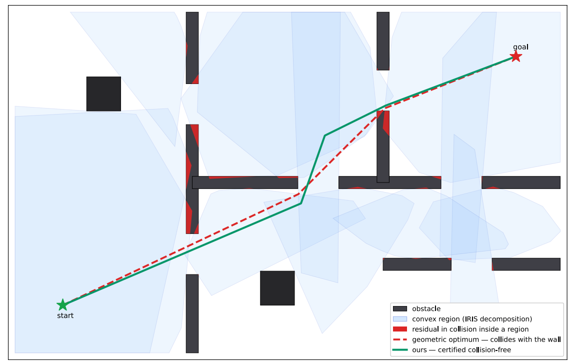

Fig. 1. Why a task fails. The decomposition tolerates thin residual wedges (red), and the reference optimum (dashed red) presses into one; our planner returns the certified path (green). Sec. III derives the mechanism.

model allows: wherever optimism shortens the path, the optimum moves there. Shortest paths also hug obstacles, exactly where a sampling-based generator's residual sits (Fig. 1), and grasp configurations put the endpoints in the same cramped geometry.

Tightening the library does not repair it: a ten times stricter acceptance contract, sums-of-squares certified regions [2], and uniform margins each dissolve the region graph before they deliver soundness (Sec. V-D).

Our planner buys certainty along the trajectory instead, and it buys it by sampling. We sample the overlaps and shared faces of the library, the places a trajectory must cross to move between regions, and check every sample against the true collision geometry before it joins the roadmap. Sampling puts the candidate set in our hands: a waypoint is a configuration that can be tested, rejected, or moved; an optimum of a convex program offers no such handle. Sampling also makes the planner fast: an informed filter discards interfaces that cannot improve the incumbent, and the convex program runs once on a short corridor. A candidate is then verified continuously, one clearance query certifying a whole ball of configurations and overlapping balls certifying the segment with no discretization step to fall through; failed segments are repaired inside the region that produced them, where convexity keeps the detour legal, and the cone program that polishes the survivor goes back through the same verification.

The result is a planner that hands back only verified answers. Across all 29 task queries the reference planner delivers 21 trajectories that fail certification and ours delivers zero, our first certified answer arrives in a fourteenth of the time the reference needs for an unverified one, and on every query

TABLE I
HOW THE REGION ASSUMPTION ENTERS FIVE REPRESENTATIVE WORKS. QUOTATIONS ARE VERBATIM.

work	treatment of "regions are collision-free"
GCS motion planning [5]	one sentence: the fast generator "does not provide a rigorous certification, but ... appears to be very reliable in practice"
GCS* [8]	guarantees stated relative to the mathematical program; regions are input
IxG* [9]	sets "capture the free and safe planning space"
Shortest walks [11]	"we produce a convex decomposition of the collision-free configuration space"
GGCS [6]	plans confined to "certified" C-Free (quotes in the original), grown by the uncertified generator

whose reference answer is physically valid we return that optimum exactly.

a) Contributions: This paper reports the first measurement of region soundness in GCS planning and of its effect on planner output, through a benchmark protocol built for bit-exact, twice-validated, cross-machine reproduction. It then gives evidence that the defect cannot be fixed in the library, since a stricter acceptance contract, sums-of-squares certificates, and uniform margins each dissolve the region graph before they make it sound. Its constructive part is a sampling-based planning layer, combining interface sampling, informed pruning, continuous clearance certificates, in-region repair, and a re-verified convex polish, that keeps the graph intact, returns certified paths or explicit failures, and matches the reference optimum wherever the reference is right. The code, both libraries, the benchmark data, and the result videos are released.

II. Related Work

This section places the paper in three bodies of work: the GCS planning family whose shared assumption we measure, the generators that produce its regions, and the checking and certificate machinery our layer builds on.

A. Planning on convex-region libraries

Marcucci et al. [4] formulate shortest paths in graphs of convex sets with a relaxation tight enough that rounded solutions are near-optimal, turned into a motion planner in [5]. A family followed: implicit search (GCS* [8]; IxG* [9]), multi-query bounds and walks [10], [11], travelling-salesman variants (GHOST [12]), tensor-train compression [7], and geodesic generalization [6]; planners over safe boxes [13] and optimized covers [14] rest on the same input.

These planners share one assumption: the regions they consume are collision-free. Table I records how five of them state it; the rest consume the decomposition as given, and none checks the returned trajectory against the collision geometry.

B. Convex region generation

IRIS [1] inflates a convex region around a seed by separating it from obstacles with hyperplanes: exact against convex workspace obstacles but not against their non-convex C-space preimages (Sec. III). Two branches answer this. In the statistical branch, IRIS-NP [23] accepts a region after a run of failed counterexample searches, a probabilistic certificate with no explicit bound, and IRIS-ZO [3] formalizes the test, bounding the colliding volume fraction by ε at confidence $1-\delta$; C-IRIS [24], [2] instead proves separation with sums-of-squares programs. The planning literature runs on the statistical branch, which builds 7- and 14-DOF libraries in minutes; the proofs cost minutes to hours per region at 7–12 DOF [2], and Sec. V-D measures what happens when they feed a planner.

C. Collision checking and feasibility certificates

A distance query returns more than a yes/no answer: the clearance at a configuration certifies a ball of collision-free configurations around it [15], and roadmaps can be built from such balls [16], also with learned configuration-distance models [17]. Lazy planners search first and validate only the path they intend to return [25], a scheme with formal edge-selection guarantees [26]. Our planner puts both to work on a convex decomposition, whose overlaps tell us exactly where a trajectory must be checked and whose convexity turns a failed segment into a repairable one. The informed anytime planners [19], [20] sample free space itself, build a fresh graph per query, and approach the optimum only asymptotically, their validity resting on resolution edge checks; we sample the decision set of a precomputed decomposition, reach the corridor optimum with one convex solve carrying a per-answer gap, and certify the continuum. In 2- and 3-D workspaces, where cells can be carved exactly, decomposition planners already win this comparison outright: neural corridor selection reports 99 to 100% success at 2 to $280\times$ the speed of sampling baselines [21], [22]. Configuration space is where the paradigm's premise breaks, the cells turning probabilistic (Sec. III); this paper carries the paradigm there with its guarantees intact.

Existing certificates describe the program (relaxation gaps, a bounded-suboptimality factor in GHOST [12]) or the regions (C-IRIS); ours describes the answer: an ε-gap on cost and a continuous clearance certificate on the path (Sec. IV-C).

III. Problem Statement

This section formalizes the gap between the modeled and the physical problem, derives why the residual must exist, and compresses its behavior into three testable predictions.

a) Setting and decomposition: Let $\mathcal{Q} \subset \mathbb{R}^d$ be the configuration space of a robot and $\mathcal{F} \subset \mathcal{Q}$ its collision-free subset, decided by a collision checker that we treat as ground truth. A query is a pair (q_s, q_g) of collision-free configurations; a path is a continuous curve $q : [0,1] \to \mathcal{Q}$ between them, with arc length as cost.

An IRIS-style generator [1] grows a convex polytope around a seed configuration by alternating two steps: inflate an ellipsoid inside the current polytope, and add a separating hyperplane, a *cut*, that pushes the polytope away from an obstacle. Grown from many seeds, the regions form a library $\mathcal{U} = \bigcup_{i=1}^{n} X_i$. Two regions are adjacent when their *interface* $X_u \cap X_v$, itself a polytope, is nonempty; the regions and

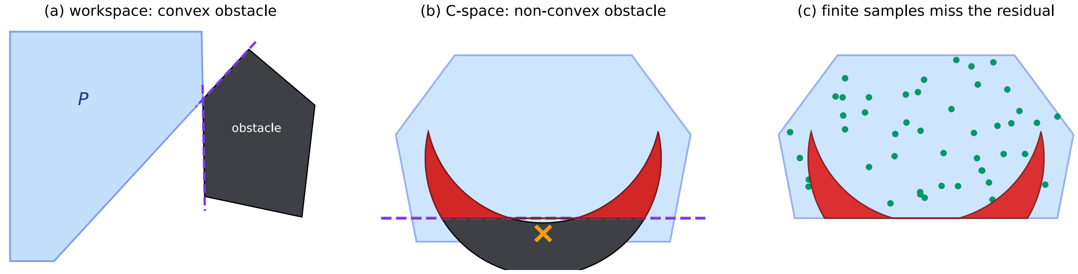

Fig. 2. Why the residual is unavoidable. (a) Against a convex obstacle, cuts placed on its own faces are a complete and exact exclusion proof: the free region hugs the obstacle. (b) The C-space obstacle is non-convex: a cut at the discovered collision ($\times$) removes a half-space, and the parts that curve back around it (red) remain inside the region. (c) Uniform samples (green) miss the thin residual, so the statistical test accepts.

interfaces form the graph the planner searches. The planner solves the *modeled* problem

$$m(\mathcal{U}) \;=\; \min\{\,\mathrm{len}(q) \;:\; q(t) \in \mathcal{U} \;\;\forall t\,\}, \tag{1}$$

and the user needs the *physical* one, with $\mathcal{U}$ replaced by $\mathcal{U}\cap\mathcal{F}$. The two coincide exactly when $\mathcal{U} \subseteq \mathcal{F}$.

b) Non-convexity and the acceptance residual: In the workspace with convex obstacles, cuts finish the proof: a convex region disjoint from a convex obstacle admits a strictly separating hyperplane, and placing the cuts on the obstacle's own faces excludes it exactly; the free region hugs the obstacle with zero slack (Fig. 2a). In configuration space the hypothesis fails. For a convex workspace body W and robot geometry $B(q)$ posed by forward kinematics, the C-obstacle is the preimage

$$O = \{\, q : \; B(q) \cap W \neq \emptyset \,\}, \tag{2}$$

and forward kinematics makes O in general non-convex. No hyperplane separates a convex region from it; a cut placed at a discovered colliding configuration removes only a supporting half-space, and the parts of O that curve back around the cut remain inside the region (Fig. 2b). Finitely many cuts cannot exclude a non-convex set in general; a universal proof needs the algebraic certificates of C-IRIS [2], whose cost is why scaled pipelines do not run it.

Unable to prove, the generator tests: accept the region after M clean uniform samples [3]. The mathematical content is the Bernoulli bound

$$\begin{aligned}&\Pr[\text{accept} \mid \text{colliding fraction} > \varepsilon] \\ &\quad \leq\; (1-\varepsilon)^M \;\leq\; e^{-\varepsilon M} \;\leq\; \delta \quad \text{for } M \geq \tfrac{\ln(1/\delta)}{\varepsilon},\end{aligned} \tag{3}$$

so acceptance certifies a colliding fraction at most ε at confidence $1-\delta$; a zero-ε contract needs $M = \infty$. Shipped defaults are $\varepsilon = 1\%$, $\delta = 5\%$, and an iteration cap after which regions are admitted with the test unfinished. We call $\mathcal{U} \setminus \mathcal{F}$ the *residual*; it is what the test tolerates: thin in volume, so the samples miss it (Fig. 2c). The residual is not a breach of the contract; it is the contract.

c) Boundary concentration and optimizer exploitation: Three effects put the tolerated volume in the planner's way. (i) Cuts are tangent at discovered collisions, so the undiscovered slivers hug the polytope faces: the residual lives in the boundary shell. (ii) High-dimensional volume concentrates

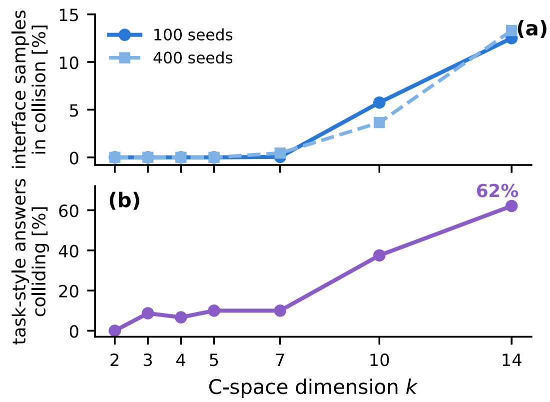

Fig. 3. Residual against C-space dimension. (a) Interface contamination on the planar k-link family (fixed workspace, obstacles, and generator; only k varies) at two seed budgets. (b) Task-style answers colliding: stress-endpoint queries on the same family through k=10; the 14-DOF point is the bimanual task benchmark (18/29). Through k=5 the volume statistic reads exactly zero while delivered answers already collide.

there: shrinking a convex body in $\mathbb{R}^d$ by 5% leaves 0.95^d of its volume, so at $d = 14$ the shell holds $1 - 0.95^{14} \approx 51\%$ of the body. (iii) Interfaces $X_u \cap X_v$ are double boundary zones, the meeting of two obstacle-hugging shells, and they are exactly where paths cross. This is why a measurement on interfaces can read 3.2% against a 1% volume contract without contradiction: it is the operationally relevant measure of the same residual.

Proposition 1. *If $m(\mathcal{U}) < m(\mathcal{U}\cap\mathcal{F})$, then every optimizer of* (1) *is physically invalid.*

Proof. A path attaining $m(\mathcal{U})$ costs less than $m(\mathcal{U}\cap\mathcal{F})$, and every path inside $\mathcal{U}\cap\mathcal{F}$ costs at least $m(\mathcal{U}\cap\mathcal{F})$. □

The hypothesis holds whenever the residual offers a shortcut, and by the concentration above it usually does: minimum-length paths press against the same shell, and grasp endpoints sit where the generator's budget runs out first.

Remark 1 (Dimension dependence of the residual)**.** *Fix the acceptance contract (ε, δ) and the seed budget. The tolerated residual keeps a volume share of at most ε in every dimension d, but effects (i)–(iii) confine it to the boundary shell of each region, and the share of a convex body within a relative*

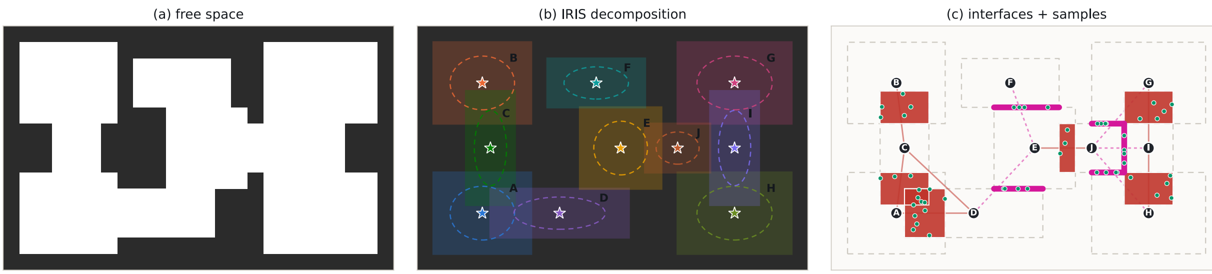

Fig. 4. From free space to a sampled decision set. (a) The free space of a floor plan. (b) Its IRIS decomposition: seeds (stars) inflate into overlapping convex regions. (c) The interfaces the decomposition induces: volume overlaps (red) and shared faces (magenta), with uniform samples (green) drawn on every interface; nodes and thin lines sketch the region graph they populate.

distance ρ of its boundary grows as $1-(1-\rho)^d$. The residual's share of the interface measure, the measure a path has to cross, therefore grows with d while the reported volume statistic does not, and by Proposition 1 an optimizer selects that residual whenever it shortens the path: the probability that a delivered answer collides rises with d even where uniform sampling reports no contamination. Fig. 3 isolates the effect on a planar k-link arm with everything but k fixed: interface contamination is zero through k=5 and 13% at k=14, while task-style answers already collide from k=3 on.

d) Predictions: Three consequences follow, and Section V tests each: the dimension dependence of Remark 1; the amplification from volume to answers, whose error exceeds the residual's volume share by an order of magnitude and grows as query endpoints approach obstacles, with Proposition 1 as its kernel; and the failure of library-level tightening, which removes interfaces before it removes the residual, because certainty about a d-dimensional volume is paid wherever the volume meets obstacles, where regions must also meet each other.

e) Certified answers: A planner consuming a library that satisfies only (3) should return, for each query, either a path q with a proof that $q(t) \in \mathcal{F}$ holds for every $t \in [0, 1]$, or an explicit report that no such path was found. The proof must cover the continuum, because the residual is thin exactly where paths run. The next section builds a planner with this property.

IV. Method: Sampling and Certification

This section builds the planner: sampling on the interfaces, informed iteration toward the optimum, and the certification layer that proves every delivered answer.

A. Interface sampling and roadmap

Inside one convex region the cheapest way between two configurations is the straight segment between them, and that segment stays inside the region. A path that visits a sequence of regions $X_{v_1}, \dots, X_{v_k}$ at minimum length is therefore piecewise linear, and its vertices lie on the interfaces $X_{v_i} \cap X_{v_{i+1}}$. The interfaces hold every free variable of the modeled problem (1), and sampling them discretizes the decision itself rather than the space around it.

Interfaces come in two shapes, both polytopes (Fig. 4). A *volume overlap* is d-dimensional: the two regions share a body of configurations, and a crossing may sit anywhere inside it. A *shared face* is $(d-1)$-dimensional: the regions meet on a facet, and every crossing lies on that facet. Contacts of lower dimension (an edge, a corner, a pinch between two obstacles) also satisfy $X_u \cap X_v \neq \emptyset$, and admitting them as transitions makes the roadmap claim passages that no robot can execute: two regions can meet at a single point wedged between obstacles, and a segment through that point is legal in the model with zero clearance in the world. We therefore require

$$\dim(X_u \cap X_v) \;\geq\; d - 1 \tag{4}$$

before an interface enters the roadmap.

Each interface is available as $\{x : A_u x \leq b_u,\ A_v x \leq b_v\}$. We detect the rows that hold with equality throughout, which gives the affine hull of the interface and its intrinsic dimension, and we sample in hull coordinates. Four samplers cover the shapes a crossing can take.

Center takes the Chebyshev center of the interface, the crossing that stays furthest from its rim. *Area* draws uniform samples in the interface, spreading the crossings over the full set of options. *Contour* samples the relative boundary, where the shortest crossing presses when a corridor turns. *Contour-in* pulls the contour inward by λ, keeping samples on the facet proper; a raw contour point sits on a lower-dimensional face, exactly the geometry that (4) rules out. All four deliver the same certified coverage on the closing benchmark, and center crossings, maximal-clearance points by construction, certify fastest; they anchor the head-to-head of Sec. V-E. Every sample is a configuration that can be tested: the collision checker keeps the clean ones, and the rest never enter the roadmap.

a) Roadmap construction: Nodes are the accepted samples together with q_s and q_g. Each node records the regions that produced it (an interface sample records both parents), and two nodes are joined when they share a region, with Euclidean distance as the edge cost. Membership is carried, not re-derived: a node inherits the regions that produced it, and duplicate nodes merge by taking the union of their region lists.

B. Informed iteration

The planner runs in rounds and improves monotonically (Fig. 5, Alg. 1). Round one samples every interface with a

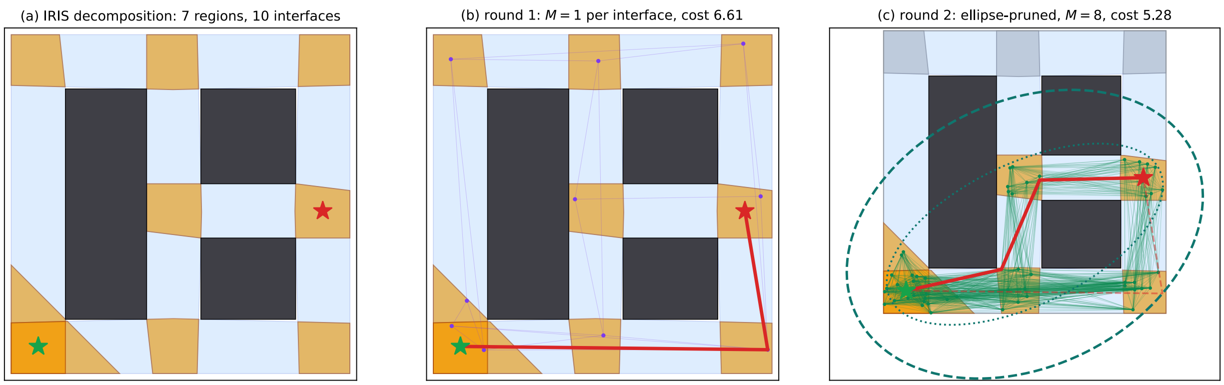

Fig. 5. The planner on a 2-D library (all elements computed). (a) IRIS decomposition; orange sets are the interfaces. (b) Round one: one sample per interface, Dijkstra on the roadmap, first incumbent. (c) The incumbent defines the informed ellipse (dashed); interfaces outside it (gray) leave the search, the survivors are resampled more densely, and the path improves; the tighter incumbent yields the smaller dotted ellipse for the next round.

Algorithm 1 Interface-sampling planner (one query)

Require: regions $\{X_i\}$, interfaces $\mathcal{I}$ with $\dim \geq d-1$, query (q_s, q_g), budget $M \leftarrow M_0$

```
c ← ∞;   A ← I                                          ▷ alive interfaces
while time remains do
    draw M samples on each F ∈ A; keep those the collision checker accepts
    roadmap ← accepted samples ∪ {q_s, q_g}; connect nodes that share a region
    repeat
        p ← shortest path on the roadmap
        p ← VerifyRepair(p)                             ▷ lazy: Sec. IV-C
        remove from the roadmap any edge it blacklisted
    until p certified or no path remains
    if p certified and len(p) < c then
        c ← len(p);   p* ← p                            ▷ certified costs drive the pruning
    end if
    A ← {F ∈ A : ℓ(F) ≤ c}                              ▷ lossless: Eq. (5)
    M ← 2M
end while
p′ ← VerifyRepair(corridor SOCP polish of p*); return the cheaper certified of p*, p′ with γ = len/LB − 1
```

small budget, searches the roadmap, and produces the first incumbent path. The incumbent's length c bounds what any better path may do. A path through interface F is at least as long as

$$\ell(F) \;=\; \min_{x \in F} \|x - q_s\| + \|x - q_g\|, \tag{5}$$

so every interface with $\ell(F) > c$, outside the informed ellipse of Gammell et al. [18] with foci q_s, q_g and major axis c, is off every improving path, and discarding it is lossless; batch and adaptive descendants [19], [20] refine the same device for sampling planners. The next round doubles the sample budget and spends it only on the survivors: the search volume shrinks as the incumbent tightens. Disabling the filter changes no answer and multiplies total planning time by 2.2; the pruning is pure speed. the denser roadmap, and the incumbent only ever improves. In Fig. 5 the first round costs 6.61; its ellipse retires the detour over the top of the map, and the resampled round returns 5.28.

Verification is interleaved, not terminal: a dirty segment is repaired by a local detour inside its own region, an unrepairable edge is blacklisted and the search reroutes within the same round, and only certified costs drive the pruning, since an unverified short path would tighten the ellipse illegally. The roadmap holds combinatorially many paths; one is checked per search, and it usually passes on the first try because its waypoints were validated before they entered the graph: one rejection across the whole curated benchmark, a single path's certification per query even at 3.2% contamination. Validating the roadmap eagerly instead costs two to three orders of magnitude more, almost all of it on edges no shortest path will ever use. The corridor polish runs once, at the end, and is itself re-verified.

C. Continuous clearance certificate

Algorithm 1 calls VerifyRepair on every candidate the search proposes; the subsections that follow implement it.

A collision checker answers more than yes or no. A clearance query at q returns $\phi(q)$, the smallest distance over all 903 filtered body pairs, measured on the same collision geometry the boolean checker uses, so the certificate and the ground truth share one oracle. Kinematics bounds how fast ϕ can fall. A point on one arm moves at $\|\dot{p}\| \leq \sum_i r_i |\dot{q}_i|$, the sum over that arm's joints, where the reach r_i is the farthest any collision geometry distal of joint i extends from its axis, and Cauchy-Schwarz turns the sum into $\sqrt{\sum_i r_i^2}\,\|\dot{q}\|$. The same constant covers arm-arm pairs, whose joint sets concatenate, and self-collision pairs, whose joint set is a subset. Hence ϕ is L-Lipschitz with $L = \sqrt{\sum_{i=1}^{14} r_i^2}$; reaches measured on the model give $L \approx 3.4$, the implementation rounds up to $L = 6.0$, and half a million random directional probes of ϕ measure a worst-case rate of 1.01, a safety factor above five. The arms carry no payload in this benchmark; a grasped object lengthens each r_i by its overhang and changes nothing else.

Lemma 1 (certificate ball)**.** *If ϕ is L-Lipschitz and $\phi(q) > 0$, then every configuration in the ball $B(q, \phi(q)/L)$ is collision-free.*

Proof. For $\|q' - q\| < \phi(q)/L$, $\phi(q') \geq \phi(q) - L\|q' - q\| > 0$. $\square$

One distance query therefore certifies a continuum of configurations, and a chain of such balls certifies a segment: starting at a, query ϕ, advance by ϕ/L along the segment, and repeat until b is reached. The balls overlap by construction, so their union covers the segment, and the segment is collision-free along its whole length. There is no resolution parameter: strides lengthen where clearance is large and shorten near obstacles, so no sliver between two test points is left for the residual. Certifying a candidate on the 14-DOF library takes about 200 clearance queries at 2.3 ms each, under half a second, mostly near obstacles where the strides shorten. The advancement stalls only where $\phi \to 0$, which is precisely a discovered violation: the certificate and the detector are the same computation. The floor $\phi_{\min} = 2 \times 10^{-3}$ that ends a stalled advancement sits on a plateau: halving or doubling it leaves the certified count unchanged.

D. In-region repair

When the advancement reports a dirty stretch on segment (a, b), the segment does not have to be abandoned. Both endpoints came from the same region X, and convexity makes X a repair kit: for any via point $z \in X$, the detour $(a, z), (z, b)$ stays inside X, so it crosses no region boundary and needs no new transitions. The repair samples candidate via points in X around the dirty stretch, tests each against the collision checker, splices in the first that certifies, and re-runs the advancement on the modified segment. The residual is thin, so a legal detour usually exists a few centimeters away; when the retry budget is exhausted the edge is blacklisted, and control returns to the search of Algorithm 1, which routes around it.

Verification produces one fact, that the candidate collides at a specific configuration q^*, and its value depends on what the planner can do with it. A convex-optimization pipeline has three options, and none of them is a repair. Re-solving returns the same optimum and the same collision; by Proposition 1, when the shortcut through the residual is real, every optimum of the program is invalid. Adding a constraint that excludes q^* moves the generator's surgery online, one supporting half-space at a time with the residual curving back around each cut (the surgical remedy of Sec. V-D). Deleting the offending region is a blacklist at the wrong granularity, a corridor for a sliver; eight of the queries our layer solves with an in-region detour have no alternative corridor.

The structural difference is ownership of the answer. An optimum is a global object that feedback reaches only by re-solving a changed problem, with no bound on the rounds. A sampled candidate is piecewise and the planner holds its waypoints: feedback lands at the failure site as a spliced via point or a blacklisted edge, the re-search is a millisecond Dijkstra, and a finite graph only shrinks.

E. Corridor polish with re-verification

The certified path is a chain of straight segments through sampled crossings, so its cost carries sampling slack. The region sequence it visits defines a corridor, and on that corridor the modeled problem (1) restricts to a second-order cone program: one free waypoint per interface, minimize total length. Solving it moves the crossings to their optimal positions and removes most of the slack: on the 14-DOF library, raw sampled paths carry $+41.5\%$ over the corridor optimum and polished paths $+9.6\%$. The polish optimizes the modeled problem, so Proposition 1 applies to it: its waypoints move toward the model's boundary, residual included. Algorithm 1 therefore sends the polished path back through VERIFYREPAIR and keeps it only when it is both certified and cheaper; otherwise the pre-polish certified path stands.

F. Output guarantees

The planner hands back a path with two guarantees of different kinds. The clearance certificate is deterministic and covers the continuum: the delivered trajectory is collision-free along its entire length. The cost certificate is the gap γ between the delivered length and a lower bound $\mathrm{LB} \leq m(\mathcal{U})$; the implementation uses the joint-space distance $\|q_s - q_g\|$, and any relaxation of (1) can tighten it. Over the fifteen certified benchmark answers γ has median 7.1%, five sit within 2%, and one attains its bound exactly.

Lemma 2 (graceful degradation)**.** *Any lower bound* $\mathrm{LB} \leq m(\mathcal{U})$ *also satisfies* $\mathrm{LB} \leq m(\mathcal{U} \cap \mathcal{F})$. *A certified path of length* ℓ *therefore satisfies* $\ell \leq (1+\gamma)\, m(\mathcal{U} \cap \mathcal{F})$ *with* $\gamma = \ell/\mathrm{LB} - 1$, *whatever the residual is.*

Proof. $\mathcal{U} \cap \mathcal{F} \subseteq \mathcal{U}$ shrinks the feasible set of (1), so $m(\mathcal{U}) \leq m(\mathcal{U} \cap \mathcal{F})$. □

Contamination widens γ; it cannot falsify it. An unvalidated geometric cost moves in the opposite direction: it quotes $m(\mathcal{U})$ as the achieved cost of a trajectory that may not be executable at all.

a) Implementation: The planner runs with M_0=1 center crossing per interface, eight doubling rounds, certificate floor $\phi_{\min}$=2×10^{-3}, L=6.0, and at most 15 via-point trials over four repair rounds per segment; the corridor polish and the reference planner's programs are solved with Clarabel through Drake, the reference in its benchmark configuration (60 s timeout, K=1, at most one revisit). All timings come from one 32-thread CPU node without GPU. The task queries are the 29 pairs of the benchmark's named grasp configurations that the library connects; the classical baseline is RRT-Connect with 0.35 rad extension, 0.03 rad edge resolution, and 60 s per seed.

V. EXPERIMENTS

This section measures the residual and tests the three predictions of Section III, runs the library-level fixes to completion, and prices the certified planner against its reference.

A. Libraries and measurement protocol

Two libraries built with the standard sampling-based generator anchor the measurement. The *curated* library covers a 7-DOF arm and a shelf with 55 regions grown from hand-picked seeds; the *scaled* library covers a 14-DOF bimanual scene (two arms, grippers, shelving, 903 filtered collision pairs) with 219 regions grown from 22 grasp seeds and 200 random seeds at the generator's default settings. Ground truth throughout is the scene's collision checker, the same oracle the generator itself sampled. The instrument: uniform interface samples tested against the oracle.

B. Interface contamination

Curation works: with seeds placed by hand and time to spare, the curated library reads 6 of 2,000 interface samples in collision (0.3%), the regime the family's demonstrations live in. Scaling changes that. Random seeds, default budgets, and an iteration cap that admits regions with the acceptance test unfinished leave 64 of 2,000 in collision (3.2%), above the per-region volume contract, exactly as the concentration argument of Section III predicts. The number is a property of the configuration, not of one draw (four regenerations at identical settings: 2.8 to 3.6%), and of dimension (zero at 2 and 3 DOF across a testbed and twelve aerial libraries, 0.15 to 0.3% at 7, 2.8 to 3.6% at 14). The controlled k-link family of Remark 1 (Fig. 3) isolates the dependence; both seed budgets trace the same law, and the optimizer finds what two thousand uniform samples cannot.

C. Answer-level failure

Volume is the cause; answers are the harm. We benchmarked the reference answers of GCS* [8] on the scaled library under a protocol built for certainty: bit-exact reproduction of the stored cost in the benchmark's exact configuration, the full waypoint sequence re-derived as the optimum of the convex program, and validation by both the continuous certificate of Section IV-C and 2,000 boolean collision probes. Of the connected task queries, 18 of 29 (62%) return colliding reference answers, against 2 of 16 random benchmark queries; violations reach 39.4 mm penetration at the median, 91.3 mm at the deepest, and one path spends 51% of its length in collision. The verdicts reproduce on a second machine and survive regeneration: four fresh instances return colliding reference answers on 83 to 84% of the connected pairs drawn from all 22 grasp configurations (38/46, 38/46, 38/46, 47/56).

Every violation is a single contiguous mid-path interval with both endpoints clear of obstacles by 38 mm or more: the grasp configurations are valid, and the path between them is what enters the shelving.

D. Library-level remedies

The measurement invites an objection: tighten the generator and the residual disappears. We ran the three natural versions of that idea to completion.

a) Stricter acceptance contract: We regenerated the scaled library from the same 222 seeds with $\varepsilon = 0.1\%$, $\delta = 1\%$, four times the particles, ten times the iterations, and the acceptance test enforced as a gate: a region that fails is regrown and finally dropped, never admitted unfinished. The gate accepts 47 of 222 seeds, only 3 of the 22 shelf-adjacent grasp seeds among them, and the survivors are pairwise disjoint: zero interfaces against 2,757 in the default library. We had registered the opposite prediction.

b) Sums-of-squares certification: C-IRIS [2] replaces the statistical test with sums-of-squares proofs. We composed it with our layer on a 2-DOF testbed, using the same seven seeds for both generators. The certificate is real: zero contamination on 3,995 benchmark samples, against a 17× generation cost. The certified regions cover 33% of free space against IrisZo's 83%, and overlap in a single pair; at 2.7 times the seed budget the certified library still fragments into seven connected components and answers none of 30 queries.

c) Uniform margins: Eroding every region by a margin δ traces the whole trade-off. Across nine margins up to $\delta = 0.10$ rad, contamination falls only from 3.39% to 2.11% while the connected benchmark queries fall from 16 to 10 of 16: the curve never approaches soundness. The residual hides in overlap volumes deeper than any uniform margin the connectivity survives.

d) : A fourth, surgical option, excising discovered residual region by region, maims the graph while failing to finish (Sec. IV-D). All four fail the same way, as Section III predicts.

E. Comparison on the scaled library

The closing experiment runs both planners on all 29 connected task queries of the 14-DOF library: same machine, same library, same queries. The reference arm is GCS* in the benchmark's exact configuration, its answer re-derived as the optimum of its own program; our arm is Algorithm 1 with center crossings. Both delivered trajectories face the same judge: the continuous certificate, with dense boolean probes as the second mechanism.

a) Correctness: The reference planner delivers 21 trajectories that fail certification (18 probe-confirmed, 3 on razor-thin margins) and reports success on every one; ours delivers 15 certified paths and 14 explicit refusals, no invalid answer. A refusal is information: every refused pair's reference answer itself fails certification, and thirteen of the fourteen carry certified witnesses from other planner runs; one defeats every planner tried. Across ten seeds the certified count moves by at most one and certified costs vary by a 0.15% median coefficient of variation.

b) Speed: Certification is not a tax; here it is a fourteen-fold speedup. Our first certified answer arrives in 0.11 s at the median against 1.59 s unverified: center crossings maximize clearance, so the first candidate's certificate balls are the largest the interfaces offer, and verification settles in about 200 clearance queries.

c) Optimality: On all eight pairs whose reference answer is physically valid, the certified cost equals the reference cost to four decimal places; the rescued pairs pay a median +9.2% (range +5.1 to +50%), the length of the legal detour around an obstacle the reference answer passes through.

d) Comparison with sampling-based planning: Narrow passages are where decomposition planners earn their keep (Sec. II-C): a measure-zero target for random sampling is an explicit interface here. RRT-Connect with the same ground-truth checker, ten seeds per pair, solves every pair in raw form, yet only 45% of its shortcut-smoothed runs survive the continuous certificate; its pipeline reaches a first certified answer in 10.7 s median against our 0.11 s, and on three pairs threading the shelf interior it certifies zero of thirty runs while the layer delivers. Where both certify, its per-seed median cost lands within 1.4% of ours: the decomposition concedes almost nothing in quality while answering deterministically, two orders of magnitude faster.

e) Curated library: Where the residual is negligible the layer is simply a fast planner: on the curated 7-DOF library the first certified incumbent arrives in 0.13 s, $21\times$ before the search baseline's best answer, and the corridor polish closes the sampling slack from $+5.7\%$ to $+1.3\%$. The supplementary video plays reference and certified answers side by side, with the carried bottles inside the collision model: a 7-DOF arm moving a bottle between shelves, where the reference sweeps hand and bottle 68 mm through a board; the bimanual arms carrying two bottles at once and handing one over, with reference violations up to 50 mm; and an aerial benchmark on which the reference optimum flies through building walls while reporting success and the certified reroute is 0.9% shorter.

VI. Conclusion

Planning on graphs of convex sets rests on regions that a sampling-based generator can only promise probabilistically, and the promise breaks precisely where optimizers look: a 3.2% interface residual on a scaled 14-DOF library leaves most task-query reference answers physically colliding, reported as successes. The cause is structural: cuts cannot exclude non-convex C-obstacles, finite samples cannot certify a continuum, and an optimizer concentrates on exactly the error the acceptance test tolerates; the library-level remedies dissolve the region graph before they deliver soundness.

The planner this paper builds accepts the library as it is and moves the guarantee to the answer. Where the model is right the guarantee costs nothing: the certified answer reproduces the reference optimum in a fourteenth of the reference's time. Where the model is wrong it returns a certified detour or an explicit refusal, never a silent violation.

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